10112 Advanced Quantum Mechanics

Hubbard model on a ring.

The Hubbard Model for two electrons on a ring: [1] This problem studies the Hubbard model in one dimension. The Hubbard model approximates the long-range Coulomb repulsion between electrons as an on-site repulsion energy U. The model is otherwise a tight-binding Hamiltonian with nearest-neighbor hopping.

The Hamiltonian is written

$$H = H_{TB} + H_C = -\sum_{n,s} t(c^\dagger_{n+1,s}c_{n,s} + c^\dagger_{n,s}c_{n+1,s}) + \sum_n U c^\dagger_{n,\uparrow}c_{n,\uparrow}c^\dagger_{n,\downarrow}c_{n,\downarrow} \tag{1}$$

n is running from 1 to L, where we imagine the L sites organized in a ring so that the electron can also hop from site L to site 1. We can see that t, the hopping matrix element, represents a kinetic energy for the electrons which is lowered if the wavefunction delocalizes into long-wavelength plane waves. Indeed, the eigenstates in the absence of electron-electron interactions are plane waves.

We can introduce the creation operator for a single-particle plane-wave:

$$\psi^\dagger_{k,s} = \frac{1}{\sqrt{L}}\sum_n e^{ink}c^\dagger_{n,s} \tag{2}$$

It can be shown (you are welcome to do that if you like) that the Hamiltonian in this representation is diagonal and equal to

$$H_{TB} = \sum_{k,s} \varepsilon_k \psi^\dagger_{k,s}\psi_{k,s}$$

with $\varepsilon_k = -2t\cos(k)$ and where k runs over $\frac{2\pi}{L}p, p = 0, 1, \ldots, L-1$.

- Discuss the meaning of this expression for H_{TB}.
- Explain what the operator H_c does to a state.

We will study the two-electron $N = 2$ problem on a ring of $L = 6$ sites, with one spin up and one spin down ($S_z = 0$). We can work in a plane-wave basis:

$$|k_1\uparrow, k_2\downarrow\rangle = \psi^\dagger_{k_1\uparrow}\psi^\dagger_{k_2\downarrow}|0\rangle \tag{3}$$

[1] This problem set was originally written by Jim Sethna at Cornell University.

The total crystal momentum is a good quantum number and we shall consider states with zero crystal momentum so $k_1 + k_2 = 0$.

- Discuss that the kinetic term H_{TB} is diagonal in this basis, and show that the on-site repulsion has matrix elements

$$\langle\, k\uparrow, -k\downarrow \,|H_C|\, k'\uparrow, -k'\downarrow \,\rangle = U/L$$

  independent of k and k'.

You now have to do a bit of numerical work. You can for example use the IPython notebook "hubbard_on_a_ring.ipynb".

- Compute the eigenfunctions and eigenenergies for total crystal momentum zero. Plot the eigenenergies vs. U for U/t ranging from zero to four.

Notice that the ground state and the highest excited state change smoothly as we increase U.

- What happens to the other two states? Which of the states are singlets, and which triplets (total spin zero and one)? Why are the triplet energies so boring?

The coulomb repulsion alters the wavefunctions rather more than is apparent in the energy.

- Calculate the ground state wavefunction in real space $\Psi_0(n_1, n_2)$. It should depend only on $n_1 - n_2$; plot $\Psi_0(n_1 - n_2)$ and discuss the hole at $n_1 = n_2$. How does the depth of the hole depend on U? Why?